HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F79SU

SURVIVAL MODELS

IN-PERSON EXAMINATION

April Examination and Assessment Diet 2025

Duration: 2 hours

Total marks available: 60

Answer all 6 questions

Attached: No additional documents attached.

Supplied: Formulae and Tables for Examinations of the Institute and Faculty of Actuaries; Distributions table.

Calculator Permitted: Yes.

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

DO NOT REMOVE THIS PAPER FROM THE EXAM HALL

1. Consider the continuous random variable T, with $T \geq 0$.

(a) Let T follow an exponential distribution, with parameter $\lambda > 0$.
Determine the hazard rate at $T = x$ for this distribution.
Hints: details of this distribution, and the distribution in part (b), can be found in the provided table of distributions. Also, the hazard rate for a continuous random variable with probability density function $f(x)$ and survival function $S(x)$ is given by $f(x)/S(x)$.

[2 marks]

(b) Let now T follow a Weibull distribution, with parameters $\nu, \lambda > 0$.
Show that the survival function for this distribution is given by

$$S(x) = \exp\left(-\lambda x^{\nu}\right).$$

Hence, determine the hazard rate at $T = x$ for this distribution.

[3 marks]

(c) Comment on the suitability of T for describing the lifetime of a policyholder in mortality studies, based on the hazard rate under each of the two distributions in parts (a) and (b). Your comments should refer in particular to the parameters of the two distributions. [4 marks]

[Total 9 marks]

2. A new drug is being developed, which doctors hope will increase the length of time between treatments for a serious disease. Data are available for a small group of patients who have either taken an old drug or the new drug, and show the time in weeks until the next treatment is required; the + indicates that the observation was censored.

Old Drug	7+	8	11	12+	12+	13	18	19
New Drug	6	10+	13+	13+	15	16+	20+	20+

Suppose that the Cox proportional hazards model holds for these data, with hazards given by $\lambda_{old}(t)$ and $\lambda_{new}(t)$ for patients taking the old and new drugs respectively. Suppose further that the group of patients taking the old drug is taken as the reference category, i.e., in the standard notation, $\lambda_{new}(t) = \lambda_{old}(t)\exp(\beta)$.

(a) Show that the partial loglikelihood $\ell(\beta)$ is [3]

$$\ell(\beta) = 2\beta - 4\log(1 + e^{\beta}) - 3\log(1 + 2e^{\beta}) \qquad (*).$$

(b) Calculate the value of $\hat{\beta}$, the maximum partial likelihood estimate of β. You do not need to check through a second derivative that your solution is a maximum of the likelihood function. [3]

(c) Given your result in part (b) alone, has the new drug performed better or worse than the old drug? Give a reason for your answer. [2]

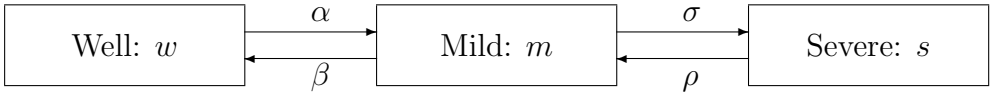
(d) Use $(*)$ to conduct the likelihood ratio test of $H_0 : \beta = 0$ *v* $H_1 : \beta \neq 0$ at the 5% level of significance and report your conclusions. [3]

[Total 11 marks]

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3. A study into sickness followed n individuals for one year from 1st January 2014 to 31st December 2014. At any given time, lives were classified as either being well, mildly sick, or severely sick. All n individuals were healthy and aged 40 at the start of the investigation. The following 3-state Markov model with constant transition intensities α, β, σ and ρ is used to model the experience of the n lives.

Note: Units are months and all months should be considered to be of equal length.



(a) Consider the experience of a life A: A becomes mildly sick on 1st April; she remains mildly sick until 1st May when she recovers; she then remains well until 1st September when she leaves the study (i.e. she is censored).

Derive, from first principles, the contribution of life A to the likelihood function for the model parameters, under the 3-state Markov model considered above.

[5 marks]

(b) The following data are available from the complete study where W_x denotes the waiting time in state x and T_{xy} denotes the number of transfers from state x to state y: $W_w = 8000$, $W_m = 1500$, $W_s = 500$, $T_{wm} = 500$, $T_{mw} = 300$, $T_{ms} = 100$ and $T_{sm} = 50$.

Note: Standard formulae may be quoted without proof.

(i) Find the maximum likelihood estimates of α, β, σ and ρ together with estimates of their standard errors, under the 3-state Markov model considered above.

[4 marks]

(ii) Use the estimates found in part (b)(i) to estimate the probability that a 40 year old individual who is well on 1st January should remain well until 31st December. [2 marks]

[Total 11 marks]

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4. The following data are available on six lives aged between x and $x+1$ where: a_i is the time in years (after age x) at which observation begins, b_i is the time in years (after age x) at which observation may be censored, and d_i indicates whether a life was censored ($d_i = 0$) or died ($d_i = 1$).

Life i	a_i	b_i	d_i
1	0	1	0
2	0	1	1
3	0.2	0.7	1
4	0.4	0.8	0
5	0.5	1	1
6	0	0.6	0

Suppose the lives are subject to a constant force of mortality $\mu = \mu_{x+t}$, for $0 \leq t \leq 1$ and let $q = q_x$.

(a) Determine the log likelihood in terms of μ for the binomial model for these data under the constant force of mortality assumption. [4]

(b) Now let $e^{-\mu} = \theta^2$. Show that the log likelihood for θ is

$$\ell(\theta) = 4\log(\theta) + 3\log(1-\theta) + \log(1+\theta).$$

Find the maximum likelihood estimate (MLE) of θ and hence deduce the MLE of q. [4]

(c) Suppose that, in addition, you are told that lives 2, 3 and 5 died at times $x+t = 0.5, 0.5$ and 0.8 respectively. Consider a 2-state Markov mortality model, with constant force of mortality, for these 6 lives.

i. Compute the exact waiting time v.

ii. Compute the maximum likelihood estimate of μ under the 2-state Markov model of mortality and deduce the maximum likelihood estimate of q.

[3]

[Total 11 marks]

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5. The table below gives the initial exposed to risk, E_x, the number of deaths, d_x, for a set of male insured lives, for ages $x = 60, \ldots, 70$. The table also gives $\text{logit}(\hat{q}_x) = \log\{\hat{q}_x/(1-\hat{q}_x)\}$, $\hat{q}_x = d_x/E_x$, and $\text{logit}(q^s_x)$ where q^s_x is taken from a standard table.

Age x	E_x	d_x	$\text{logit}(\hat{q}_x)$	$\text{logit}(q^s_x)$
60	66833	447	-5.001	-4.312
61	60135	449	-4.890	-4.196
62	56272	464	-4.790	-4.079
63	51980	481	-4.673	-3.960
64	47395	488	-4.566	-3.842
65	30007	333	-4.490	-3.728
66	20062	241	-4.410	-3.619
67	17182	248	-4.224	-3.514
68	15332	262	-4.052	-3.412
69	13570	242	-4.009	-3.315
70	11769	256	-3.806	-3.222

An actuary decides to graduate the data by using the values from the standard table and the following graduation:

$$\text{logit}(\dot{q}_x) = 1.19\,\text{logit}(q^s_x)$$

(a) Use this graduation to:

- estimate the fitted mortality rate $\dot{q}_{60}$;
- estimate the fitted number of deaths $\dot{d}_{60}$;
- calculate the standardised residual z_{60}.

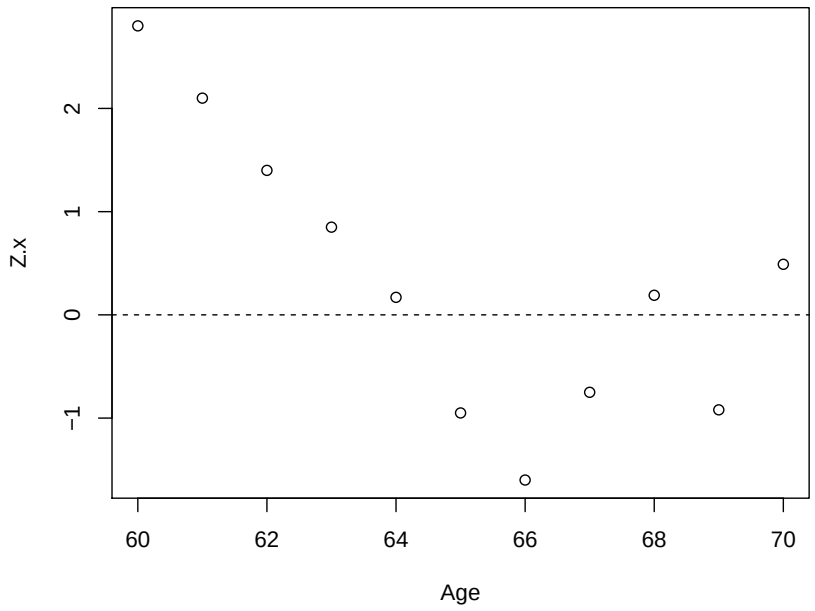
[4]

(b) Figure 1 gives a plot of the standardised residuals from this graduation. Use the plot to carry out the following tests of the graduation at the 5% level of significance: the sign test, the change of sign test and the grouping of signs test.
Hint: You may use the table of critical values for the grouping of signs test. [6]

(c) What is your general conclusion based on Figure 1 and the results of your tests in part (b)? [2]

[Total 12 marks]

Figure 1: Standardised residuals z_x against age for the graduation $\text{logit}(\dot{q}_x) = 1.19\,\text{logit}(q^s_x)$.



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6. The following numbers of deaths have been observed at age 80 *last* birthday during the calendar years 2013, 2014 and 2015.

2013	2014	2015
13	10	9

Also available are the following census data at age x *nearest* birthday, on 1 January in each of 2013, 2014, 2015 and 2016.

Age	1/1/2013	1/1/2014	1/1/2015	1/1/2016
79	532	523	522	566
80	489	501	512	499
81	524	514	503	493

(a) Find from the data in the table the number of lives aged 80 at 1/1/2013 according to the definition of age for deaths, justifying your answer and stating any assumptions you make.

[2 marks]

(b) You are given that an estimate of the central exposure to risk for age 80, from these data, is $E^c_{80} = 1516.25$.

Calculate an estimate for μ_{80} and explain the exact age at which this estimate would apply.

[2 marks]

(c) Using your answer in part (b), estimate q_{80} and state the exact age to which this estimator would apply.

[2 marks]

[Total 6 marks]

[Grand total: 60 marks]

END OF PAPER

Some Continuous Distributions

Name	Notation	p.d.f.	Range	m.g.f.	Mean	Variance
Uniform	$\mathrm{U}(a,b)$	$\frac{1}{b-a}$	$a \le x \le b$	$\frac{e^{bt}-e^{at}}{(b-a)t}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	$\mathrm{N}(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	$-\infty < x < \infty$	$e^{\mu t+\frac{1}{2}\sigma^2 t^2}$	μ	σ^2
Log Normal	$\mathrm{LogN}(\mu,\sigma^2)$	$\frac{1}{x\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(\ln x-\mu)^2}{2\sigma^2}\right\}$	$0 < x < \infty$	undefined for $t > 0$	$e^{\mu+\frac{1}{2}\sigma^2}$	$(e^{\sigma^2}-1)(e^{2\mu+\sigma^2})$
Gamma	$\mathrm{Gamma}(\alpha,\lambda)$	$\frac{\lambda^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\lambda x}$	$0<x<\infty$, $\alpha,\lambda>0$	$\left(\frac{\lambda}{\lambda-t}\right)^\alpha$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
Exponential	$\mathrm{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$0<x<\infty$, $\lambda>0$	$\frac{\lambda}{\lambda-t}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Chi-square	χ^2_ν	$\frac{1}{2^{\nu/2}\Gamma(\nu/2)}x^{\nu/2-1}e^{-x/2}$	$0<x<\infty$, $\nu>0$ integer	$(1-2t)^{-\nu/2}$	ν	2ν
Beta	$\mathrm{Beta}(\alpha,\beta)$	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}$	$0<x<1$, $\alpha,\beta>0$	*omitted	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Cauchy	$\mathrm{Cauchy}(\mu,\sigma)$	$\frac{\sigma}{\pi(\sigma^2+(x-\mu)^2)}$	$\sigma > 0$	does not exist		
Pareto	$\mathrm{Pareto}(k,\theta)$	$\frac{k}{\theta}\left(1+\frac{x}{\theta}\right)^{-(k+1)}$	$x>0$, $\theta,k>0$	does not exist	$\frac{\theta}{k-1}$	$\frac{\theta^2 k}{(k-2)(k-1)^2}$
Weibull	$\mathrm{Weibull}(\nu,\lambda)$	$\nu\lambda x^{\nu-1}\exp\left(-\lambda x^\nu\right)$	$x>0$, $\nu,\lambda>0$	*omitted	$\lambda^{-1/\nu}\Gamma\left(1+\frac{1}{\nu}\right)$	*omitted
Multivariate Normal	$\mathrm{N}_p(\boldsymbol{\mu},\boldsymbol{\Sigma})$	$\frac{\exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}}{(2\pi)^{p/2}\det(\boldsymbol{\Sigma})^{1/2}}$	$\mathbf{x}\in\mathbb{R}^p$	$e^{\boldsymbol{\mu}^{\mathrm{T}}\mathbf{t}+\frac{1}{2}\mathbf{t}^{\mathrm{T}}\boldsymbol{\Sigma}\mathbf{t}}$	$\boldsymbol{\mu}$	covariance matrix $\boldsymbol{\Sigma}$

Some Discrete Distributions

Name	Notation	p.m.f.	Range	m.g.f.	Mean	Variance
Discrete Uniform	$\mathrm{DU}(a,b)$	$\frac{1}{b-a+1}$	$x = a,a+1,\ldots,b-1,b$ a,b integers	$\frac{e^{at}-e^{(b+1)t}}{(b-a+1)(1-e^t)}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2-1}{12}$
Binomial	$\mathrm{Bin}(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	$x = 0,1,2,\ldots,n$ $0<p<1$	$[(1-p)+pe^t]^n$	np	$np(1-p)$
Poisson	$\mathrm{Poi}(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$x = 0,1,2,\ldots$ $\lambda>0$ real	$e^{\lambda(e^t-1)}$	λ	λ
Hyper-geometric	$\mathrm{HypGeo}(n,K,N)$	$\frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$	$x = 0,1,2,\ldots,n$ N,K,n integers	*omitted	$\frac{nK}{N}$	$\frac{nK(N-K)(N-n)}{(N)^2(N-1)}$
Geometric	$\mathrm{Geo}(p)$	$p(1-p)^{x-1}$	$x = 1,2,\ldots$ $0<p<1$	$p[e^{-t}-(1-p)]^{-1}$	$1/p$	$(1-p)/p^2$
Negative Binomial	$\mathrm{NegBin}(k,p)$	$\binom{x-1}{k-1}p^k(1-p)^{x-k}$	$x = k,k+1,\ldots$ $0<p<1$	$p^k[e^{-t}-(1-p)]^{-k}$	k/p	$k(1-p)/p^2$

*omitted due to lack of space

Multinomial: multivariate extension of binomial

Probability mass function (p.m.f) is given by

$$f(x_1,x_2,\ldots,x_k)=\frac{n!}{x_1!x_2!\ldots x_k!}p_1^{x_1}p_2^{x_2}\ldots p_k^{x_k},\qquad 0<p_i<1,\forall i=1,2,\ldots,k$$

where $p_1+p_2+\cdots+p_k=1$ and $x_1+x_2+\cdots+x_k=n$.

The m.g.f. is given by

$$[p_1e^{t_1}+p_2e^{t_2}+\cdots+p_ke^{t_k}]^n.$$

Moments: $\mathbb{E}(X_i)=np_i$, $\mathbb{V}\mathrm{ar}(X_i)=np_i(1-p_i)$, $\mathbb{C}ov(X_i,X_j)=-np_ip_j$, $i,j=1,2,\ldots,n$.